GRADE 12 CHEMISTRY

UNIT 1 — ACID–BASE EQUILIBRIA

Acid–base theories, aqueous equilibria, buffers, salt hydrolysis and titration

Exam-focused revision booklet

Prepared from the uploaded Ethiopian Grade 12 Chemistry Student Textbook (Ministry of Education, 2023). The explanations below are newly written study notes, not copied textbook pages. Diagrams are original schematic summaries.

How to use this booklet

Read the core idea, memorize the key relationship or equation, redraw the visual summary from memory, then answer both alternative questions without looking. For calculations, show units and working. For process questions, give the stages in the correct order.

CORE CONCEPTS	KEY EQUATIONS / PROCESSES	EXAM PRACTICE

Coverage: 19 focused topic sheets plus this guide page = 20 pages minimum. The two questions on each topic are alternatives for self-testing.

1.01 Arrhenius acids and bases

Core concept

An Arrhenius acid increases the concentration of H■ (more accurately H■O■) in aqueous solution. An Arrhenius base increases OH■ in water. Strong acids and bases ionize almost completely; the textbook examples include HCl, HNO■, H■SO■ and NaOH. The definition is restricted to aqueous solutions and does not describe every proton-transfer reaction.

Remember

HClO■ + H■O → H■O■ + ClO■■ | NaOH → Na■ + OH■

Arrhenius acids and bases	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define an Arrhenius acid and base, using one equation for each.

B. State two limitations of the Arrhenius model and classify NaCl(aq).

1.01 • Exam diagram & MCQ practice

Key exam idea: In water, an Arrhenius acid increases H_3O^+ concentration, while an Arrhenius base increases OH^- concentration.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which substance is an Arrhenius base in aqueous solution?

A. NaOH
B. HCl
C. NaCl
D. CH_3COOH

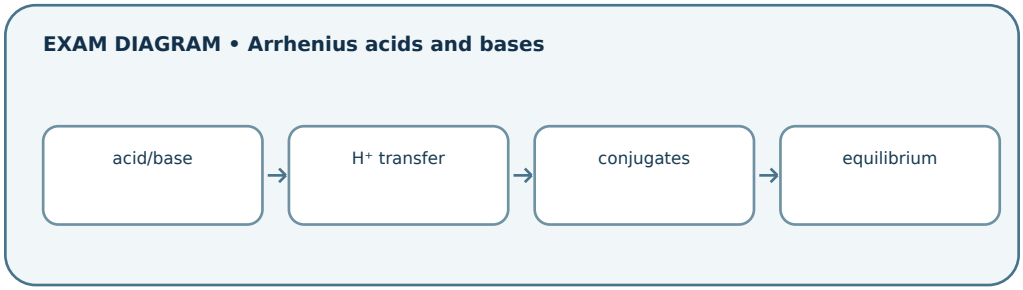
Answer: A. NaOH dissociates in water to produce OH^- ions. HCl and CH_3COOH are acids, while NaCl is a salt.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.02 Brønsted–Lowry theory

Core concept

A Brønsted–Lowry acid is a proton donor; a base is a proton acceptor. The definition focuses on transfer of H■ and is not limited to aqueous solutions. In NH■ + H■O ■ NH■■ + OH■, water donates a proton and ammonia accepts it. Identify the proton transfer before assigning acid/base labels.

Remember

Acid = H■ donor; base = H■ acceptor

Brønsted–Lowry theory	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. In NH■ + H■O ■ NH■■ + OH■, identify the acid and base.

B. Explain why NH■ is a Brønsted–Lowry base but is not an Arrhenius hydroxide.

1.02 • Exam diagram & MCQ practice

Key exam idea: A Brønsted–Lowry acid donates a proton (H^+), and a base accepts a proton.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. In $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, which species accepts a proton?

A. NH_3
B. H_2O
C. NH_4^+
D. OH^-

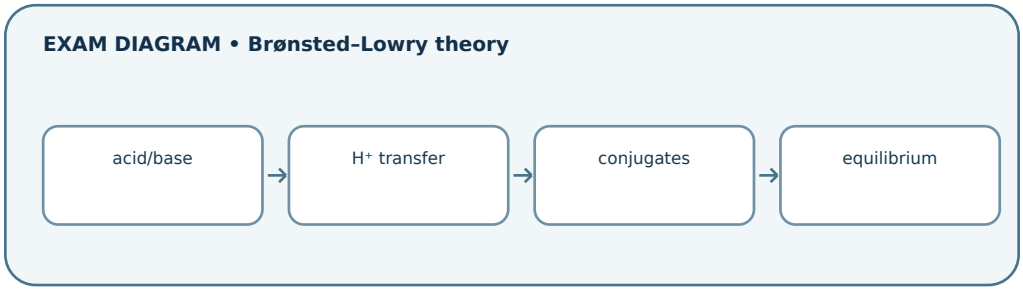
Answer: A. NH_3 accepts H^+ from water and becomes NH_4^+, so NH_3 is the Brønsted–Lowry base.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.03 Conjugate acid–base pairs

Core concept

A conjugate base remains after an acid loses one proton. A conjugate acid forms when a base gains one proton. Members of a pair differ by exactly one H■. In CH■COOH + H■O ■ CH■COO■ + H■O■, the pairs are CH■COOH/CH■COO■ and H■O■/H■O. The stronger an acid, the weaker its conjugate base.

Remember

HA ■ H■ + A■; B + H■ ■ BH■

Conjugate acid–base pairs	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Identify both conjugate pairs in HF + H■O ■ F■ + H■O■.

B. For H■PO■■ + OH■ ■ HPO■²■ + H■O, label all four species.

1.03 • Exam diagram & MCQ practice

Key exam idea: Conjugate acid–base members differ by exactly one proton.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. In $HF + H_2O \rightleftharpoons F^- + H_3O^+$, which pair is a conjugate acid–base pair?

A. HF / F^-
B. HF / H_3O^+
C. H_2O / F^-
D. F^- / H_2O

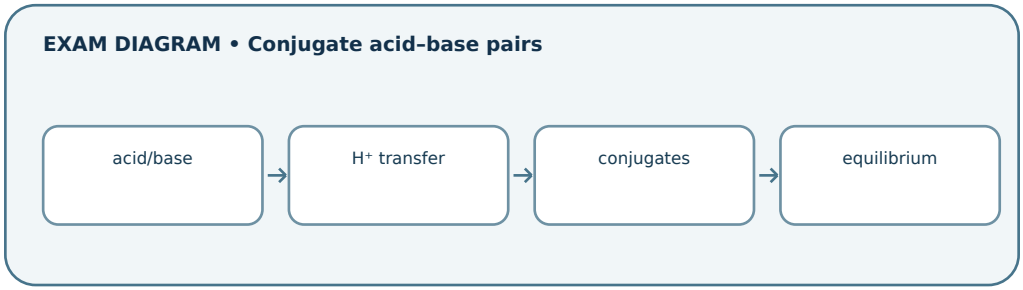
Answer: A. HF loses one proton to form F^-. Therefore HF and F^- are a conjugate acid–base pair.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.04 Relative strength of conjugate pairs

Core concept

Acid–base reactions tend to proceed in the direction that forms the weaker acid and weaker base. A strong acid has a very weak conjugate base, while a strong base has a very weak conjugate acid. Compare strengths by considering how readily a species donates or accepts a proton, not simply by counting hydrogen atoms.

Remember

Stronger acid ■ weaker conjugate base

Relative strength of conjugate p	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain why Cl■ is a weak base when it is the conjugate base of HCl.

B. Predict the favored direction when a strong acid and strong base react to form weaker conjugates.

1.04 • Exam diagram & MCQ practice

Key exam idea: The stronger an acid, the weaker its conjugate base; equilibrium favors formation of the weaker acid–base pair.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. The conjugate base of a strong acid is generally:

A. very weak
B. very strong
C. always neutral
D. unable to exist

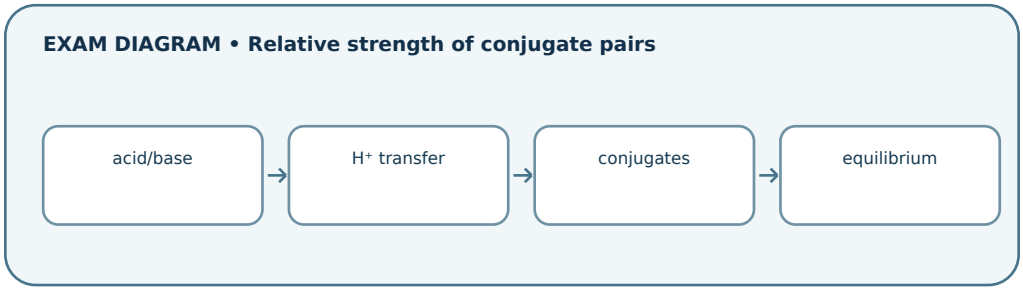
Answer: A. A strong acid readily donates H^+, so its conjugate base has little tendency to accept H^+.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.05 Autoionization and amphiprotic species

Core concept

Autoionization is a reaction between identical molecules that produces ions. Water undergoes a very small degree of self-ionization. An amphiprotic species can either donate or accept a proton depending on its reaction partner. Water and HCO■■ are examples. This dual behavior explains why water can act as either acid or base.

Remember

2H■O ■ H■O■ + OH■

Autoionization and amphiprotic s	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define autoionization and write the self-ionization equation of water.

B. Show one reaction in which HCO■■ acts as an acid and one in which it acts as a base.

1.05 • Exam diagram & MCQ practice

Key exam idea: Water self-ionizes: $2H_2O \rightleftharpoons H_3O^+ + OH^-$. An amphiprotic species can donate or accept a proton.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which equation represents the autoionization of water?

A. $2H_2O \rightleftharpoons H_3O^+ + OH^-$
B. $H_2O \rightarrow H_2 + O_2$
C. $H_2O + O_2 \rightarrow H_2O_2$
D. $H_3O^+ + OH^- \rightarrow H_2$

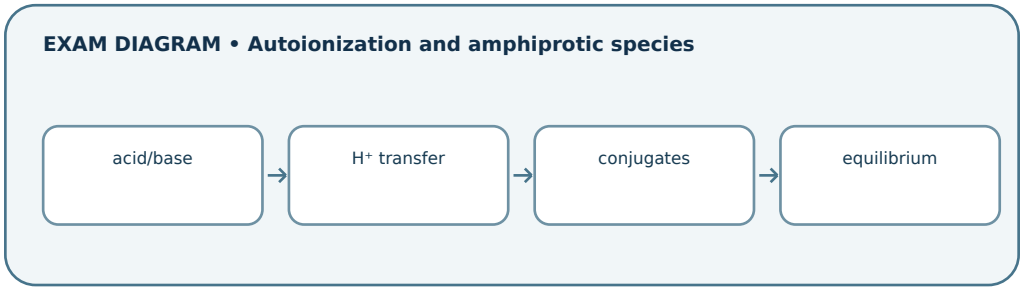
Answer: A. Two water molecules exchange a proton, producing hydronium and hydroxide ions.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.06 Lewis acid–base concept

Core concept

A Lewis acid accepts an electron pair to form a bond; a Lewis base donates an electron pair. This model includes reactions that do not involve proton transfer. In BF■ + NH■, electron-deficient boron accepts a lone pair from nitrogen. Look for an electron-pair donor and an electron-pair acceptor.

Remember

BF■ + :NH■ → F■B←NH■

Lewis acid–base concept	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Identify the Lewis acid and base in BF■ + NH■.

B. Explain why BCl■ is a Lewis acid when it reacts with Cl■.

1.06 • Exam diagram & MCQ practice

Key exam idea: A Lewis acid accepts an electron pair; a Lewis base donates an electron pair.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. In $BF_3 + NH_3 \rightarrow F_3B{\leftarrow}NH_3$, which is the Lewis acid?

A. BF_3
B. NH_3
C. F^-
D. H_2O

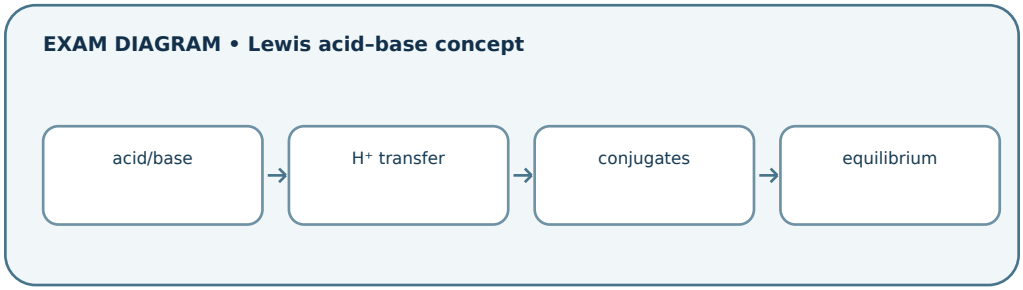
Answer: A. BF_3 accepts a lone pair from NH_3 to form a coordinate covalent bond.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.07 Water ion product, Kw

Core concept

The self-ionization of water establishes an equilibrium between H■O■ and OH■. At 25°C, Kw = [H■O■][OH■] = 1.0 × 10■¹■. In pure water the ion concentrations are equal, so each is 1.0 × 10■■ M. Kw changes with temperature; use the value and temperature stated in a question.

Remember

Kw = [H■O■][OH■] = 1.0 × 10■¹■ at 25°C

Water ion product, Kw	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Calculate [OH■] when [H■O■] = 2.0 × 10■■ M at 25°C.

B. Find [H■O■] in pure water at 25°C and explain why the solution is neutral.

1.07 • Exam diagram & MCQ practice

Key exam idea: At 25 °C, Kw = $[H_3O^+][OH^-] = 1.0 \times 10^{-14}$.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. At 25 °C, if $[H_3O^+] = 1.0 \times 10^{-5}$ M, what is $[OH^-]$?

A. 1.0×10^{-9} M
B. 1.0×10^{-5} M
C. 1.0×10^{-14} M
D. 1.0×10^{9} M

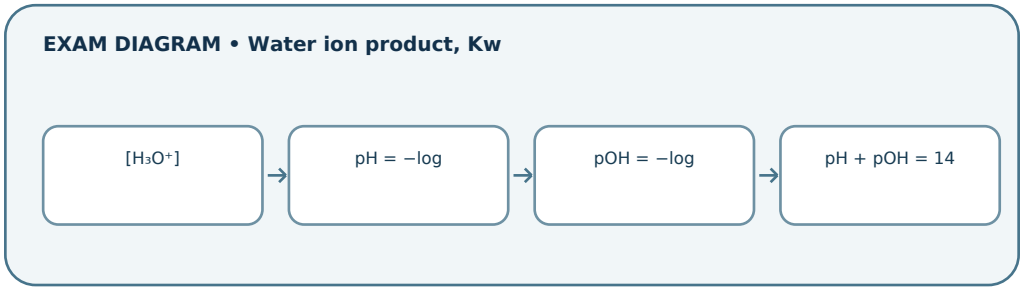
Answer: A. Use $[OH^-] = Kw/[H_3O^+] = 10^{-14}/10^{-5} = 10^{-9}$ M.

Question 2. In the diagram, which labelled step comes immediately after "$[H_3O^+]$"?

A. pH = −log
B. pOH = −log
C. pH + pOH = 14
D. $[H_3O^+]$

Answer: A. The schematic is arranged in sequence: $[H_3O^+]$ → pH = −log → pOH = −log → pH + pOH = 14.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.08 pH, pOH and concentration

Core concept

pH is the negative logarithm of hydronium concentration; pOH is the negative logarithm of hydroxide concentration. At 25°C, pH + pOH = 14. A neutral solution has equal [H■O■] and [OH■]; acidic solutions have more H■O■, while basic solutions have more OH■. Convert carefully between logarithmic values and concentrations.

Remember

pH = –log[H■O■]; pOH = –log[OH■]; [H■O■] = 10■pH

pH, pOH and concentration	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Calculate pH and pOH for [H■O■] = 5.0 × 10■³ M.

B. A solution has pH 9.18. Calculate [H■O■], pOH and [OH■].

1.08 • Exam diagram & MCQ practice

Key exam idea: pH = −log[H_3O^+], pOH = −log[OH^-], and at 25 °C pH + pOH = 14.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A solution has [H_3O^+] = 1.0 × 10^{-3} M. Its pH is:

A. 3
B. 11
C. 7
D. −3

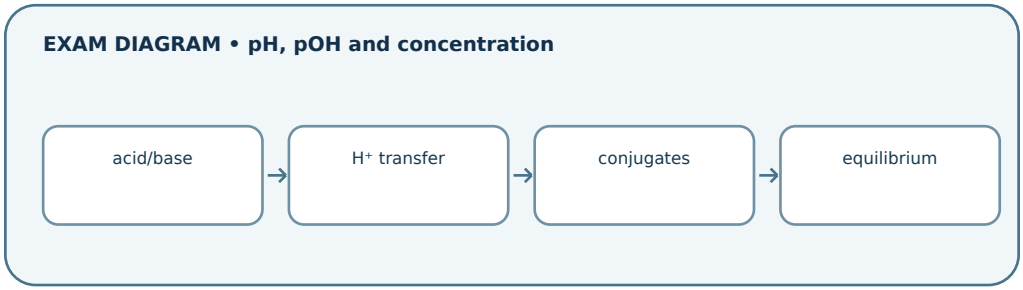
Answer: A. pH = −log(1.0 × 10^{-3}) = 3. The solution is acidic.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.09 Strong and weak acids/bases

Core concept

Strong acids and bases ionize almost completely in water; weak acids and bases ionize only partially and establish equilibrium. Strength is not the same as concentration: strength describes extent of ionization, while concentration describes amount per volume. For equal concentrations, a stronger acid generally produces more hydronium ions.

Remember

Strong: near-complete ionization; weak: partial ionization

Strong and weak acids/bases	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Distinguish acid strength from acid concentration.

B. Why can a dilute strong acid be less concentrated but still be classified as strong?

1.09 • Exam diagram & MCQ practice

Key exam idea: Strong acids and bases ionize almost completely; weak acids and bases ionize only partially.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which statement correctly distinguishes strength from concentration?

A. Strength describes extent of ionization
B. Strength is the amount of solute per litre
C. A dilute acid must be weak
D. A concentrated acid must be strong

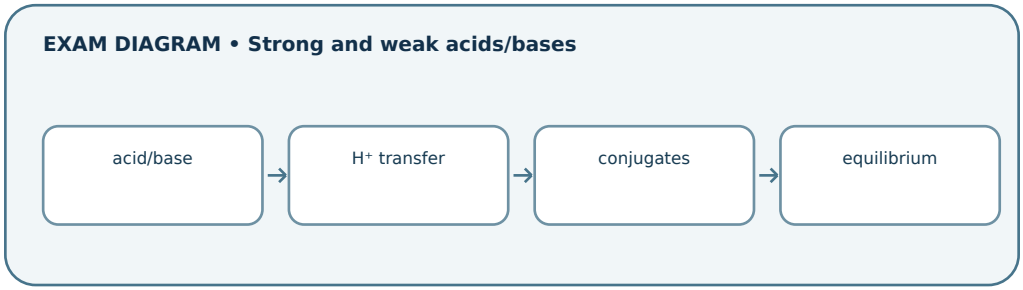
Answer: A. Strength concerns ionization in water; concentration measures the amount of dissolved substance per volume.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.10 Percent ionization

Core concept

Percent ionization measures the fraction of the initial acid or base that has ionized at equilibrium. A larger percentage indicates greater ionization under the stated conditions. Determine the equilibrium amount ionized, divide by the initial concentration, and multiply by 100%. Do not use the initial concentration in the numerator.

Remember

% ionization = (equilibrium ionized concentration ÷ initial concentration) × 100%

Percent ionization	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. A weak acid starts at 0.20 M and ionizes by 0.0040 M. Find its percent ionization.

B. Explain how percent ionization changes when comparing stronger and weaker acids of equal initial concentration.

1.10 • Exam diagram & MCQ practice

Key exam idea: Percent ionization = (amount ionized ÷ initial amount) × 100%.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A 0.20 M acid ionizes by 0.010 M. What is its percent ionization?

A. 5.0%
B. 0.5%
C. 20%
D. 50%

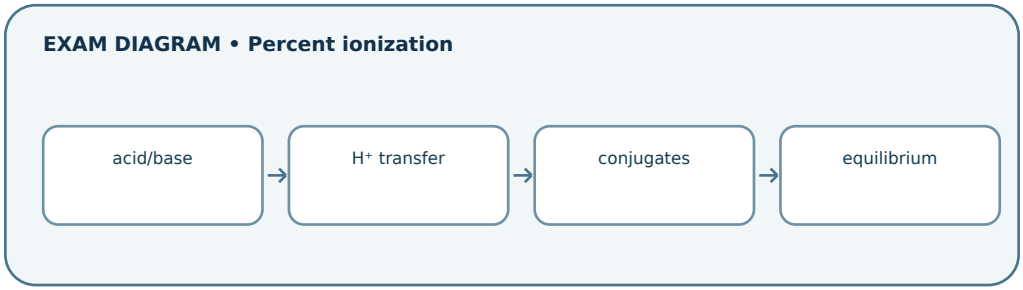
Answer: A. (0.010/0.20) × 100 = 5.0%.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.11 Acid dissociation constant, Ka

Core concept

For HA + H■O ■ H■O■ + A■, the acid dissociation expression omits liquid water because its concentration is effectively constant. A larger Ka means a stronger weak acid relative to another weak acid at the same temperature. Use an ICE table when equilibrium concentrations are unknown.

Remember

Ka = [H■O■][A■]/[HA]

Acid dissociation constant, Ka	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the Ka expression for HCOOH in water.

B. A 0.250 M weak acid has pH 2.72. Outline how to calculate its Ka.

1.11 • Exam diagram & MCQ practice

Key exam idea: For HA + H_2O ⇌ H_3O^+ + A^-, Ka = $[H_3O^+][A^-]/[HA]$.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is the correct Ka expression for HA in water?

A. $[H_3O^+][A^-]/[HA]$
B. $[HA]/([H_3O^+][A^-])$
C. $[H_3O^+][HA]/[A^-]$
D. $[A^-]/[HA]$

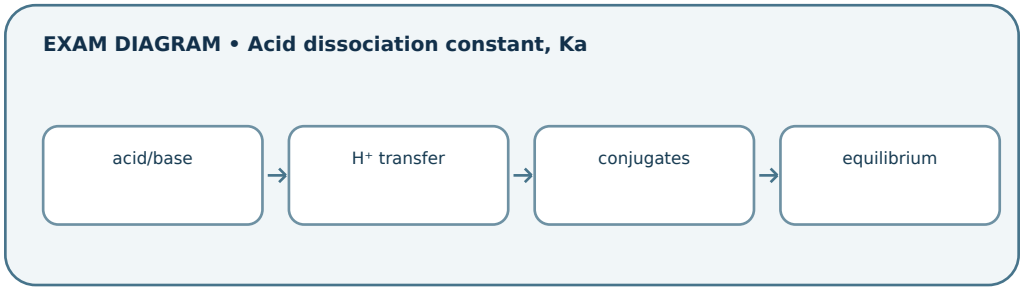
Answer: A. The equilibrium expression places product concentrations over reactant concentrations; pure liquid water is omitted.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.12 Base dissociation constant, Kb

Core concept

A weak base reacts with water to produce its conjugate acid and OH■. Kb is the equilibrium constant for this ionization. A larger Kb indicates a stronger base. Set up initial, change and equilibrium concentrations, then substitute into the Kb expression. The textbook treats pOH and hydroxide concentration as useful measures of base strength.

Remember

B + H■O ■ BH■ + OH■; Kb = [BH■][OH■]/[B]

Base dissociation constant, Kb	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the Kb expression for NH■ reacting with water.

B. Explain how Kb indicates relative base strength.

1.12 • Exam diagram & MCQ practice

Key exam idea: For B + H_2O ⇌ BH^+ + OH^-, Kb = $[BH^+][OH^-]/[B]$.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which product is formed when a Brønsted base B accepts a proton from water?

A. BH^+
B. B^-
C. H_3O^+
D. H_2

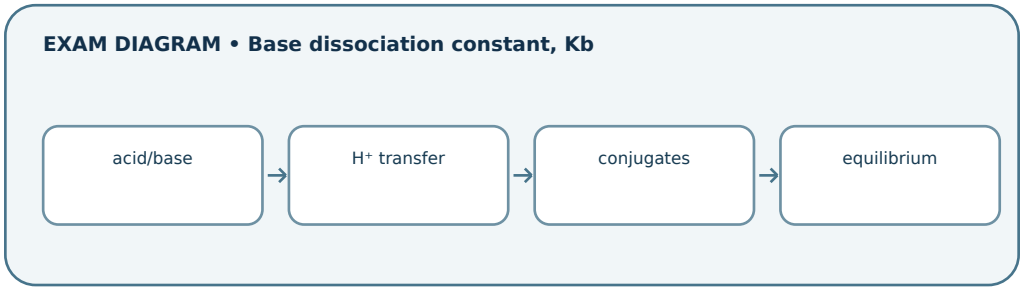
Answer: A. The base accepts H^+ and becomes its conjugate acid BH^+; water loses H^+ and forms OH^-.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.13 Weak acid equilibrium calculations

Core concept

For a monoprotic weak acid HA, let x be the amount ionized. At equilibrium [H■O■] = x, [A■] = x and [HA] = C■ – x. If x is small compared with C■, an approximation may simplify the calculation, but check its validity. If pH is given, first calculate x = 10■pH.

Remember

Ka = x²/(C■ – x)

Weak acid equilibrium calculatio	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Set up an ICE table for a 0.10 M weak acid with Ka = 1.8 × 10■■.

B. A weak acid has pH 3.00. Determine the hydronium concentration before calculating Ka.

1.13 • Exam diagram & MCQ practice

Key exam idea: For a monoprotic weak acid of initial concentration C, an ICE table gives $[H_3O^+] = x$ and $[A^-] = x$ at equilibrium.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. For $HA \rightleftharpoons H^+ + A^-$, if x is the amount ionized from an initial concentration C, the equilibrium [HA] is:

A. $C - x$
B. $C + x$
C. $x - C$
D. C/x

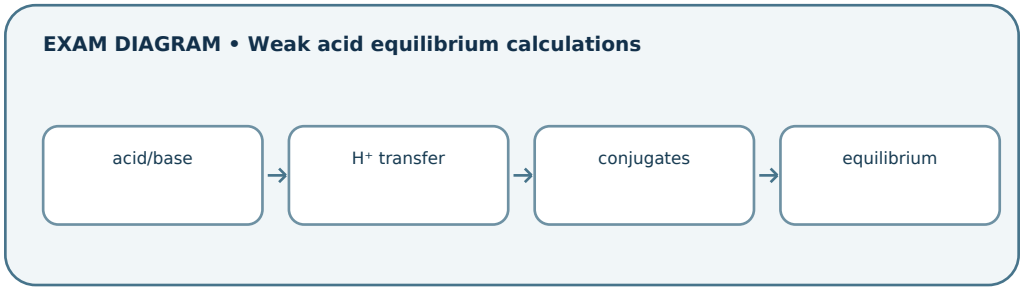
Answer: A. The acid concentration decreases by the amount that ionizes, leaving $C - x$.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.14 Weak bases and polyprotic acids

Core concept

Weak bases establish equilibrium with water and generate OH■. Polyprotic acids can donate more than one proton in successive stages; each stage has its own dissociation equilibrium. The first ionization is commonly treated separately from later ones. Track charges and hydrogen atoms carefully in each step.

Remember

H■A ■ H■ + HA■; HA■ ■ H■ + A²■

Weak bases and polyprotic acids	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the first two ionization steps of H■CO■.

B. Explain why a diprotic acid has more than one acid dissociation constant.

1.14 • Exam diagram & MCQ practice

Key exam idea: A polyprotic acid can donate more than one proton in successive ionization steps, each with its own Ka.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A diprotic acid can donate:

A. two protons per molecule
B. one proton per molecule only
C. three electrons
D. two hydroxide ions

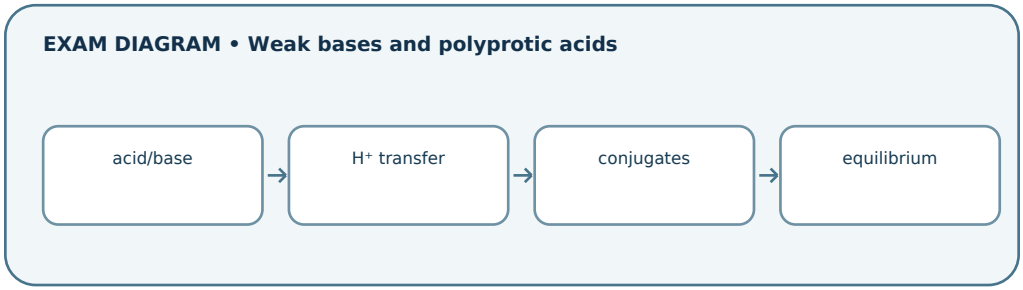
Answer: A. The prefix di- indicates two ionizable protons, usually released in separate equilibrium steps.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.15 Common-ion effect

Core concept

Adding an ion already present in an equilibrium shifts the equilibrium to reduce the disturbance. For a weak acid, adding its conjugate-base ion suppresses further ionization. This is an application of Le Châtelier's principle and is important in buffer preparation and controlling solubility or ion concentrations.

Remember

HA ■ H■ + A■; adding A■ shifts left

Common-ion effect	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Predict the effect of adding sodium acetate to acetic acid solution.

B. Explain why adding NH■Cl reduces the ionization of NH■ in water.

1.15 • Exam diagram & MCQ practice

Key exam idea: Adding a common ion shifts an equilibrium to oppose the increase, often suppressing ionization of a weak electrolyte.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Adding sodium acetate to acetic acid solution generally:

A. decreases acetic acid ionization
B. increases acetic acid ionization
C. converts all acid to water
D. has no equilibrium effect

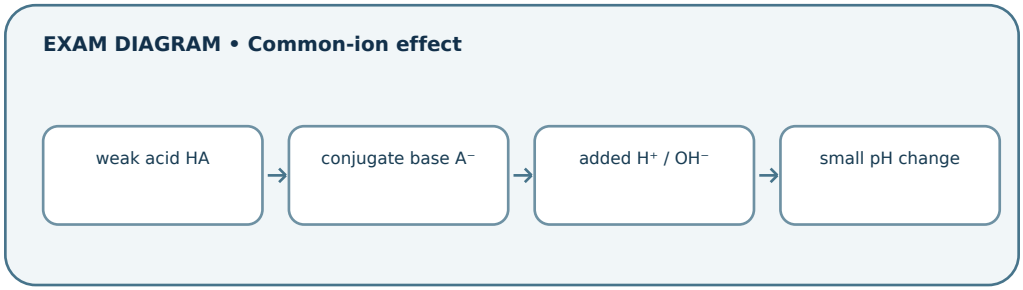
Answer: A. Acetate is a common ion. Its addition shifts the acid ionization equilibrium toward undissociated acetic acid.

Question 2. In the diagram, which labelled step comes immediately after “weak acid HA”?

A. conjugate base A^-
B. added H^+ / OH^-
C. small pH change
D. weak acid HA

Answer: A. The schematic is arranged in sequence: weak acid HA → conjugate base A^- → added H^+ / OH^- → small pH change.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.16 Buffer solutions

Core concept

A buffer resists large pH changes when small amounts of acid or base are added. An acidic buffer commonly contains a weak acid and its conjugate base; a basic buffer contains a weak base and its conjugate acid. The pair consumes added H■ or OH■, limiting pH change. Buffer capacity is limited by the amounts present.

Remember

Weak acid/conjugate base pair; weak base/conjugate acid pair

Buffer solutions	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Identify a suitable conjugate pair for an acidic buffer.

B. Explain how a buffer responds separately to a small addition of HCl and NaOH.

1.16 • Exam diagram & MCQ practice

Key exam idea: A buffer contains a weak acid and its conjugate base, or a weak base and its conjugate acid, and resists pH change.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which mixture can form an acidic buffer?

A. CH_3COOH and CH_3COONa
B. HCl and NaCl
C. NaOH and NaCl
D. HNO_3 and KNO_3

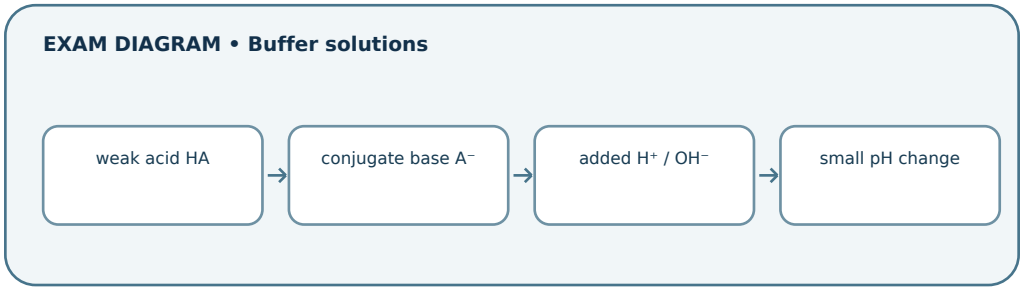
Answer: A. A weak acid and its conjugate base are present in the acetic acid/acetate mixture.

Question 2. In the diagram, which labelled step comes immediately after “weak acid HA”?

A. conjugate base A^-
B. added H^+ / OH^-
C. small pH change
D. weak acid HA

Answer: A. The schematic is arranged in sequence: weak acid HA → conjugate base A^- → added H^+ / OH^- → small pH change.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.17 Salt hydrolysis

Core concept

Salt ions may react with water and alter pH. A salt from a strong acid and strong base generally gives a neutral solution. A salt from a weak acid and strong base tends to give a basic solution; a salt from a strong acid and weak base tends to give an acidic solution. For a salt from a weak acid and weak base, compare the relevant Ka and Kb values.

Remember

Strong acid + strong base salt → approximately neutral

Salt hydrolysis	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Predict whether NH■Cl and CH■COONa solutions are acidic, basic or neutral.

B. Explain why NaCl solution is approximately neutral.

1.17 • Exam diagram & MCQ practice

Key exam idea: Salt ions may react with water; salts of a strong acid and strong base are generally neutral, while other combinations may be acidic or basic.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. An aqueous solution of NH_4Cl is generally:

A. acidic
B. basic
C. neutral
D. always pH 14

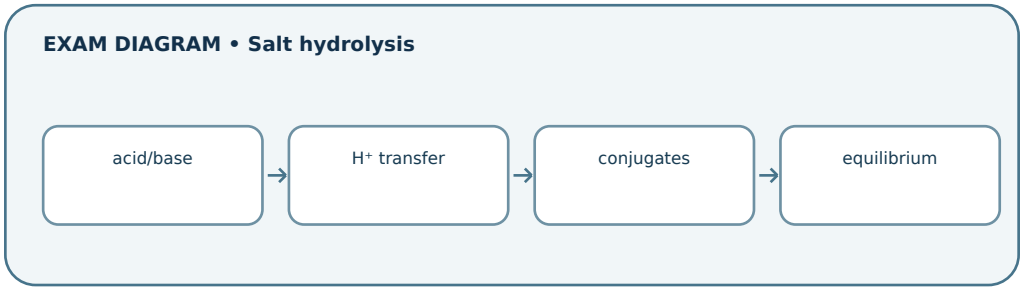
Answer: A. NH_4^+ is the conjugate acid of the weak base NH_3 and reacts with water to produce H_3O^+.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.18 Acid–base indicators

Core concept

Indicators are weak acids or bases whose forms have different colours. The observed colour depends on the solution pH and the indicator transition range. Choose an indicator whose colour-change range overlaps the steep pH change near the titration equivalence point. Common examples discussed include litmus, methyl orange and phenolphthalein.

Remember

HIn ■ H■ + In■ (two differently coloured forms)

Acid–base indicators	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Why is indicator choice important in an acid–base titration?

B. Select a suitable indicator for a strong acid–strong base titration and justify your choice.

1.18 • Exam diagram & MCQ practice

Key exam idea: An indicator has differently coloured forms; its colour change range should overlap the steep pH change near the equivalence point.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Why must an indicator be selected carefully for a titration?

A. Its transition range should match the steep pH change
B. It must react completely with the analyte
C. It should always be colourless
D. It determines the solution volume

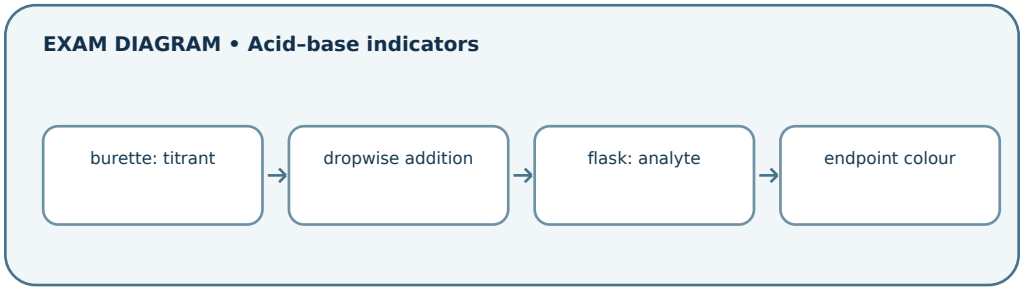
Answer: A. A suitable indicator changes colour close to the equivalence point, giving a reliable endpoint.

Question 2. In the diagram, which labelled step comes immediately after “burette: titrant”?

A. dropwise addition
B. flask: analyte
C. endpoint colour
D. burette: titrant

Answer: A. The schematic is arranged in sequence: burette: titrant → dropwise addition → flask: analyte → endpoint colour.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.19 Equivalents and neutralization

Core concept

An equivalent relates the amount of acid or base to the amount that reacts in a neutralization. The number of replaceable H■ ions per acid molecule or OH■ ions per base formula unit affects equivalent amount. In calculations, use the balanced equation and the reacting capacity of each substance rather than assuming every acid is monoprotic.

Remember

At equivalence: acid equivalents = base equivalents

Equivalents and neutralization	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Determine the neutralizing capacity of one mole of H■SO■.

B. How many moles of NaOH are needed to neutralize 0.10 mol H■SO■?

1.19 • Exam diagram & MCQ practice

Key exam idea: Neutralization depends on reacting equivalents; one mole of H_2SO_4 can supply two moles of H^+.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. How many moles of NaOH are needed to neutralize 0.10 mol H_2SO_4 completely?

A. 0.20 mol
B. 0.10 mol
C. 0.05 mol
D. 0.30 mol

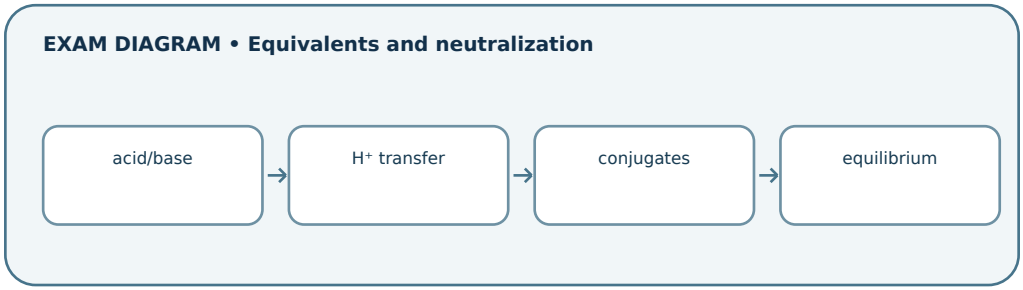
Answer: A. H_2SO_4 is diprotic: $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$.

Question 2. In the diagram, which labelled step comes immediately after “acid/base”?

A. H^+ transfer
B. conjugates
C. equilibrium
D. acid/base

Answer: A. The schematic is arranged in sequence: acid/base → H^+ transfer → conjugates → equilibrium.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



1.20 Acid–base titration

Core concept

A titration uses a solution of known concentration to determine an unknown concentration. The equivalence point is where stoichiometric amounts have reacted; the end point is the indicator colour change. Write and balance the reaction, use mole ratios, and distinguish equivalence point from end point. Titration curves show pH change as titrant is added.

Remember

For 1:1 reaction: M■V■ = M■V■; otherwise use balanced mole ratio

Acid–base titration	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Describe the main steps in an acid–base titration.

B. 25.0 mL acid is neutralized by 20.0 mL of 0.10 M NaOH in a 1:1 reaction. Find acid molarity.

1.20 • Exam diagram & MCQ practice

Key exam idea: A titration uses a standard solution to determine an unknown concentration; at equivalence, reacting amounts are stoichiometrically equivalent.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. 25.0 mL of acid is neutralized by 20.0 mL of 0.10 M NaOH in a 1:1 reaction. What is the acid concentration?

A. 0.080 M
B. 0.125 M
C. 0.20 M
D. 0.040 M

Answer: A. $M_aV_a = M_\beta V_\beta$, so $M_a = (0.10 \times 20.0)/25.0 = 0.080$ M.

Question 2. In the diagram, which labelled step comes immediately after “burette: titrant”?

A. dropwise addition
B. flask: analyte
C. endpoint colour
D. burette: titrant

Answer: A. The schematic is arranged in sequence: burette: titrant → dropwise addition → flask: analyte → endpoint colour.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.

